

OutageIntel AI: Project Documentation

Telecom Incident Knowledge Hub

1. Project Overview

OutageIntel AI is an advanced incident management and knowledge platform tailored for telecommunications operators. The system monitors, analyzes, and assists in the resolution of network outages by transforming raw telemetry and logs into actionable engineering intelligence.

2. Problem Statement

Current telecom monitoring ecosystems often flood Network Operation Centers (NOC) with alarms without providing historical context or automated Root Cause Analysis (RCA). This leads to increased Mean Time to Repair (MTTR) and difficulty in preventing recurring systemic failures.

3. Objectives

- **Intelligent Analysis:** Automate Root Cause Analysis using LLMs.
- **Trend Identification:** Detect failure hotspots and equipment degradation patterns.
- **Knowledge Retention:** Build a centralized hub for historical incident resolution.

- **Scalability:** Utilize FastAPI and Supabase for real-time high-concurrency data handling.

4. Core Technologies

Technology	Functional Purpose
Python	Backend processing, network telemetry parsing, and AI orchestration.
FastAPI	High-performance RESTful API development for incident reporting.
Supabase	Cloud-based PostgreSQL database for incident logs and metadata.
Groq / LLM	Natural Language Generation for RCA and Post-Incident Reviews (PIR).

5. Key Features

Incident Lifecycle Management

End-to-end tracking of network faults from initial alarm trigger to final resolution and documentation.

AI-Powered Root Cause Analysis (RCA)

Utilizes Large Language Models to interpret technical logs and suggest the most likely cause of failure (e.g., fiber cut, power surge, or hardware failure).

Predictive Risk Scoring

Identifies network nodes showing signs of instability before a total outage occurs.

6. Application Scenarios

Scenario: Regional Fiber Cut

A backhaul link is severed during construction. OutageIntel AI correlates downstream site failures, identifies the precise break location via telemetry, and suggests rerouting protocols to the engineering team within seconds.

7. Project Workflow

1. **Milestone 1:** System Design & Telecom Schema Definition.
2. **Milestone 2:** Telemetry Ingestion & Database Integration.
3. **Milestone 3:** AI Engine Development for RCA & Summarization.
4. **Milestone 4:** Dashboard Visualization & Deployment.